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ORIGINAL COMMUNICATION

Neural drive increases following resistance training in patients with multiple sclerosis

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Abstract The present study tested the hypothesis that lower body progressive resistance training (PRT) increases the neural drive expressed as surface electromyographical (EMG) activity in patients with multiple sclerosis (MS). The study was a randomised controlled trial (RCT) including a 12-week follow up period. Thirty-eight MS patients were randomized to an exercise group ($n = 19$) or a control group ($n = 19$). During the intervention period, the exercise group performed a 12-week supervised lower body PRT program, whereas the control group maintained their usual daily activity level. After the 12 week intervention period, the exercise group were encouraged to continue training on their own for a 12-week follow up period, while the control group completed the 12-week supervised PRT program. Surface EMG was recorded from vastus lateralis, rectus femoris and semitendinosus during maximal isometric knee extension and knee flexion, before and after the intervention and at follow up. From the recordings, the area under the rectified, low-pass filtered EMG signal (integrated EMG, iEMG) was calculated. Muscle strength was expressed as the angular impulse (AI) and was measured during the same period as the iEMG. After PRT significant improvements for iEMG of vastus lateralis and rectus femoris during maximal knee extension and for semitendinosus during maximal knee flexion and for AI during both maximal knee extension and flexion were found in the exercise group, when compared to the control group. When compared to the post values, all effects, except for AI during knee flexion, were maintained at follow up in the exercise group. When the control group was exposed to PRT, a similar pattern of improvements were found, albeit not all improvements were significant. In conclusion twelve weeks of intense PRT of the lower extremities improved the neural drive expressed as maximal surface EMG activity in patients with MS, with effects persisting 12 weeks after the intervention. The study was registered at clinicalTrials.gov, Protocol no. NCT00381576.

Keywords Exercise · Muscle strength · Strength training · Exercise therapy · MS rehabilitation

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Introduction

Multiple sclerosis (MS) is a clinically and pathologically complex and heterogeneous disorder of unknown etiology in the central nervous system [16]. Many MS patients are characterized by impaired functional capacity [26], which is related to reduced muscle strength [32] affecting predominantly the lower extremity [27]. The mechanisms underlying this strength deficit are probably of both muscular [4, 14, 17] and neural [8, 23] origin. A loss of muscle mass is seen in some studies [12, 14, 17], but not in all [4, 19, 23], along with reduced muscle fiber size and minor changes in the distribution of fiber types [4, 17]. Regarding neural mechanisms

underlying the observed strength deficit, a recent study by Scott et al. [28] showed that neural drive, assessed by surface EMG, was reduced in moderately impaired MS patients (EDSS 4–6), when compared to healthy, matched controls. In addition, muscle activation during maximal voluntary contractions is reduced in MS patients (47–93 % activation) as compared to healthy controls (94–100 % activation) [8, 23, 24, 29]. Also, decreases in maximal motor unit firing frequencies have been reported in MS patients [24].

We have previously demonstrated that 12 weeks of progressive resistance training (PRT) improves isometric muscle strength and functional capacity in moderately impaired MS patients, where the isometric peak torque of knee extensors and knee flexors increased by 16 and 21 %, respectively [5]. From healthy subjects it is further known, that PRT improves neural function in terms of improved recruitment and activation of the involved muscles (for review please see Folland and Williams [11]).

It is important to develop interventions that can improve neural function or delay the degenerative process in patients with a neurodegenerative disease like MS. Nevertheless, a recent review showed that only a few studies have investigated the neural effects of PRT in MS patients [18]. In one of these studies, White et al. [33] found no change in voluntary quadriceps femoris activation after 8 weeks of PRT applying the superimposed twitch technique in a non-controlled study including 8 MS patients. In contrast, Fimland et al. [10] reported in a randomized controlled trial (RCT), that the maximal neural drive to the soleus muscle (measured by surface EMG) increased by 40 %, when compared to the non-training control group, after only 3 weeks of PRT. However, this study applied a training frequency of 5 days a week, which would be difficult to implement in most patients' daily life. Furthermore, neural drive of some of the prime movers (knee flexors and knee extensors) of the lower limb was not evaluated. In summary, the existing studies are few and inconsistent and the neural effects of PRT on the prime movers need to be further clarified in RCT studies applying a lower training frequency that can be implemented easier in daily living. Consequently, this study tested the hypothesis, that PRT improves neural drive during maximal isometric contractions in persons with MS. Furthermore, we wanted to characterize baseline associations between surface EMG, muscle strength and anthropometric measures, primarily to establish the influence of skinfold thickness on the surface EMG signal, but also to evaluate whether muscle strength and surface EMG are closely related in patients with MS.

Methods and materials

The present study was a two-arm, randomized controlled trial lasting in total 24 weeks (12-week trial period followed by a 12-week follow up period—see Fig. 1). During the 12-week trial period, the exercise group performed a supervised, lower body PRT program, whereas the control group maintained their usual daily activity level. During the subsequent 12-week follow up period, the exercise group was encouraged to continue training on their own (without any supervision or facilities being offered), while the control group completed the same supervised PRT program initially completed by the exercise group. Participants were tested before (pre, week 0) and after (post, week 12) the trial period, and again at follow up (follow up, week 24).

Subjects

Of 426 screened patients with MS, 106 patients fulfilled the inclusion criteria: definite relapsing–remitting MS according to the McDonald criteria [21], Expanded Disability Status Scores between 3.0 and 5.5 with a pyramidal function score ≥2.0, ability to walk ≥100 m, no need for help with transportation to training facility and age >18 years with accept of diagnosis and treatment. Patients were excluded if they suffered from dementia, alcoholism or had a pacemaker implanted, had any serious medical co-morbidities, had had an MS attack within the last 8 weeks, were pregnant or had participated in systematic PRT within the last 3 months. During the study, the participants were excluded if they had an attack influencing the pyramidal functions or if they participated in less than 80 % of the training sessions. From the pool of 106 potential participants, 38 patients were included via the out-patient MS Clinic at Aarhus University Hospital during a 17-month period (November 2, 2006, to March 31, 2008), and randomly assigned, with respect to gender, to either the exercise group ($n = 19$) or the control group ($n = 19$). Eligible patients ($n = 106$) were contacted in a random order until the required number of patients were included. Of the 95 patients contacted, 36 did not respond to the invitation and 21 rejected participation. A total of 31 participants (Exercise: completers $n = 15$; drop outs $n = 4$; Control: completers $n = 16$; drop outs $n = 3$) completed the study and were included in the final data analysis of the longitudinal data. Except for one subject who experienced worsening of lower back pain, all other drop-outs were caused by circumstances unrelated to the intervention (for a detailed flowchart please see Dalgas et al. [6]). The number of recruited patients was based on a power analysis, where muscle strength was the primary outcome (expected group difference = 16 %; SD = 16 %; $p = 0.05$; desired power = 0.8; expected dropout rate = 15 %). The calculation was conducted in STATA (StataCorp LP, College Station, TX).

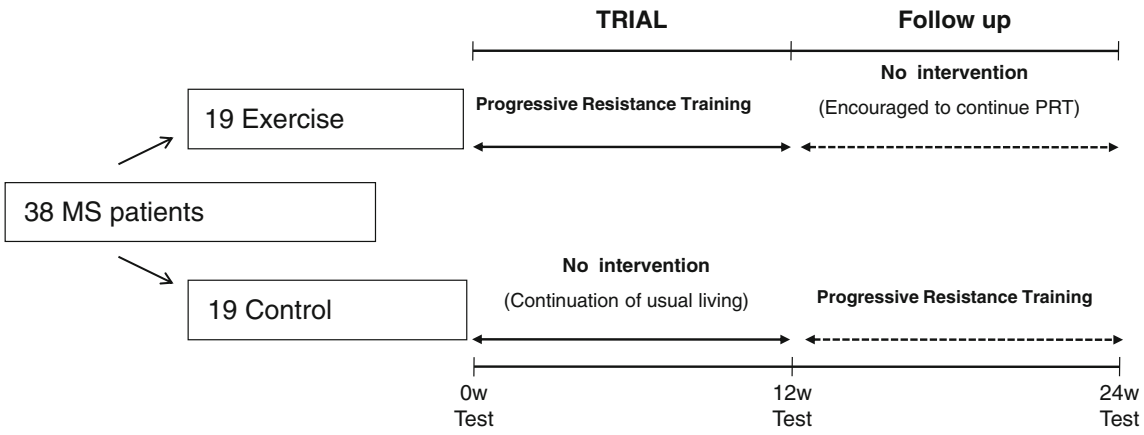

Fig. 1 Study design

Standard protocol approvals, registrations, and patient consents

The study was approved by the local scientific ethics committee (Videnskabsetisk Komité, Aarhus Amt, journal no. 20060088) and performed in accordance with the Helsinki Declaration 2. Written, informed consent was obtained from all participating patients and the study was registered at http://www.clinicaltrial.gov (NCT00381576).

Progressive resistance training

The PRT intervention consisted of 12 weeks of resistance training of the lower extremities performed twice weekly (Monday and Thursday). After a 5-min warm-up on a stationary bicycle, patients performed five different exercises: leg press, knee extension, hip flexion, hamstring curl, and hip extension. The participants were instructed to perform all exercises at a fast concentric pace and a slow eccentric pace. In terms of sets, repetitions, and load, the progression model was as follows: weeks 1 and 2, 3 sets of 10 repetitions at a load of 15 repetitions maximum (RM); weeks 3 and 4, 3 sets of 12 repetitions at a load of 12RM; weeks 5 and 6, 4 sets of 12 repetitions at a load of 12RM; weeks 7 and 8, 4 sets of 10 repetitions at a load of 10RM; weeks 9 and 10, 4 sets of 8 repetitions at a load of 8RM; weeks 11 and 12, 3 sets of 8 repetitions at a load of 8RM. Between sets and exercises, a rest period of approximately 2–3 min was allowed. To ensure progression, all training sessions were supervised. Most of the training sessions were conducted in groups of 2–4 subjects. If a participant missed a training session, it was attempted to substitute the session on an alternate day.

Maximum isometric muscle strength

Maximum voluntary contraction (MVC) of the knee extensors and the knee flexors were measured with a Biodex System 3 PRO dynamometer (Biodex Medical System, Shirley, New York). The dynamometer used for testing was not applied during the training sessions, minimizing the effects of learning. The better functioning lower leg (patient reported) was attached to the lever arm two cm proximal to the medial malleolus, and the axes of the lever arm and the knee joint were aligned. During the following maximal muscle contractions, arms were crossed over the chest and the upper body was strapped to the chair. The individual fitting of the dynamometer chair and lever arm were registered and used in all three test-sessions. Isometric muscle strength was tested at 70° knee flexion (0° being fully extended). The participants were asked to perform each attempt as hard and fast as possible. At first, the participants attempted to maximally extend their knee for approximately 4 s followed by a 1-min rest. Then they attempted to maximally flex the knee for another 4 s followed by a 1-min rest. This sequence was repeated three times with torque being sampled at 100 Hz.

To obtain a strength measure from the same time interval as the iEMG recordings (see EMG measurements), the angular impulse (AI) was calculated for 3½ s after onset of contraction during both maximal knee extensions and flexions. Data analysis was performed using custom-made software built in LabVIEW 2011 (National Instruments, Austin Texas, USA). The torque signal was low-pass filtered using a 4th order, zero lag Butterworth filter with a cut off frequency of 8 Hz. Onset of contraction was detected by a custom-made algorithm calculating a moving average with a window length of 0.6 ms and comparing this to the value just to the right of the window. In case of a deviation exceeding 3 % onset was defined. The AI measure was calculated as the average score of attempt number 2 and 3. The first attempt for all subjects was excluded from analysis since none of the participants demonstrated their best muscle strength performance during this attempt.

Testing was performed by the principal investigator who was not blinded to the intervention.

EMG measurements

Surface EMG signals[1] were recorded during maximal isometric testing from vastus lateralis (VL), rectus femoris (RF) and semitendinosus (ST), using bipolar surface electrodes (Ambu Blue Sensor, Medicotest A/S, Ballerup, Denmark) applying a 2 cm interelectrode distance and having a 0.6 cm^2 recording area. After careful preparation of the skin (shaving, abrasion and cleaning with alcohol), the upper electrode of each pair was placed at a level corresponding to the midpoint between spina iliaca anterior superior and the patellar basis. The exact electrode positions were carefully measured, and a map showing both the electrode positions and recognisable marks on the thigh was drawn on transparent paper. A Viking Select System, version 9 (Viasys Healthcare Neurocare Group, San Diego, USA) was used for the EMG recordings. The sampling frequency was 100 Hz. After data collection, further data analysis was performed using custom-made software built with LabVIEW 2011 (National Instruments, Austin Texas, USA). The raw EMG signal was full-wave rectified and low-pass filtered with a 4th order Butterworth filter with 6 Hz cut-off frequency, resulting in a linear envelope, from which the area under the curve (integrated EMG; iEMG) was calculated from onset of force contraction and 3.5 s ahead (see Fig. 2). To allow direct comparison to the AI measure, iEMG was calculated as the average iEMG score of attempt number 2 and 3. Co-contraction of vastus lateralis and rectus femoris during knee flexion and of semitendinosus during knee extension was calculated as the antagonist-to-agonist ratio and expressed as a percentage of maximal agonist iEMG, here exemplified by vastus lateralis (VL) pre-test values:

$$\text{Co-contraction VL-iEMG}_{\text{pre}} = (\text{VL-iEMG knee flexion}_{\text{pre}} / \text{VL-iEMG knee extension}_{\text{pre}} \times 100).$$

In accordance with earlier studies [1], the iEMG during maximal isometric contractions was interpreted as the maximal neural drive.

Thigh volume and skinfold thickness

The anthropometric technique for estimating total thigh volume (TV_{Total}) was applied as previously described [15].

To evaluate possible changes in subcutaneous thigh fat during the intervention, Harpenden calliper skinfold measurements (Baty International, West Sussex, England) were taken over the vastus lateralis and the rectus femoris muscles at a level corresponding to the midpoint of a line connecting the patellar basis to the midpoint between the pubic tubercle and the anterior superior iliac spine. To allow direct comparison to iEMG and muscle strength data, only data from the best functioning leg was included in this report. All anthropometric testing was done by a blinded investigator.

Statistics

Normal distribution of data was checked with box-plots, q–q-plots, histograms and dot-plots.

Pairwise correlation analysis was conducted on baseline data from all participants ($n = 38$) to evaluate relevant associations between measures of iEMG, isometric muscle force, and skinfolds of the best functioning thigh. Correlations were categorized as weak ($r = 0.30$–0.50), moderate ($r = 0.51$–0.70), high ($r = 0.71$–0.90) or very high ($r > 0.90$).

As described elsewhere [5], the primary outcome measures of the study were knee extensor MVC and a functional capacity score, while all other measures were secondary outcomes. However, the main focus of this part of the study was on changes in iEMG and AI in the exercise group vs. the control group during the 12 week intervention period [($\text{Exercise}_{\text{post}} - \text{Exercise}_{\text{pre}}$) vs. ($\text{Control}_{\text{post}} - \text{Control}_{\text{pre}}$)]. Secondary outcome measures were changes in iEMG and AI at 12-week follow up ($\text{Exercise}_{\text{post}}$ vs. $\text{Exercise}_{\text{Follow up}}$) as well as changes in iEMG and AI in the control group after 12 weeks of normal daily activity compared to 12 weeks of PRT[($\text{Control}_{\text{post}} - \text{Control}_{\text{pre}}$) vs. ($\text{Control}_{\text{followup}} - \text{Control}_{\text{post}}$)].

Since group changes were defined as the primary measure of interest these were initially compared by a two way repeated measures ANOVA, followed by the Student–Newman–Keuls post hoc test, if allowed. To compare Pre values and time and group effects, a further supplementary two way repeated measures ANOVA was performed again followed by the Student–Newman–Keuls post hoc test when relevant. Categorical data (EDSS score) were compared by a Wilcoxon signed rank test. A further pairwise correlation analysis was conducted on changes of iEMG and AI. Changes from both groups ($n = 31$) during the PRT intervention were pooled and included in the analysis. All outcome measures were tested statistically using a 5 % limit of significance. Data are presented as mean ± SD in tables and mean ± SE in figures. Statistical analyses were performed using STATA version 11 (StataCorp, Texas) and Sigmaplot 11 (Sysstat software, London).

Results

Subject characteristics and adherence

Table 1 shows subject characteristics and information about disease severity, duration and treatment of MS

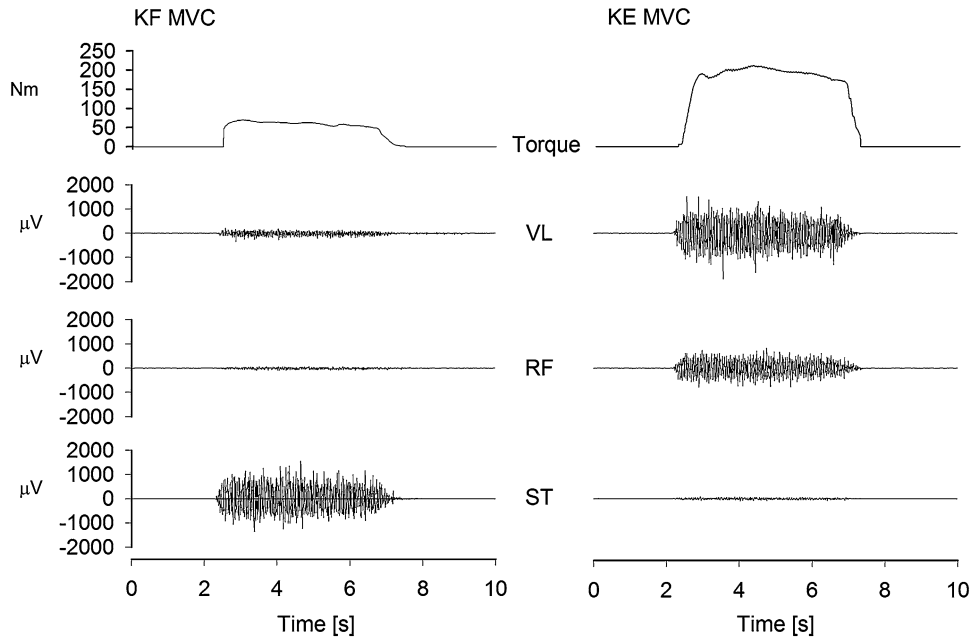

Fig. 2 Typical recording of the isometric knee joint torque and surface EMG of vastus lateralis (*VL*), rectus femoris (*RF*) and semitendinousus (*ST*) during **a** maximal knee flexion (KF-MVC) and **b** maximal knee extension (KE-MVC)

patients. The male:female ratio of all the participants was approximately 1:2. No differences were present between the exercise and the control groups before the intervention. EDSS and body weight remained unchanged during the trial in both groups. Participants in the Exercise group completed a total of 23.9 ± 0.3 out of the planned 24 PRT sessions. After the trial, participants in the control group completed an average of 22.6 ± 2.1 out of 24 planned PRT sessions during their 12 weeks exercise period.

Baseline correlations between iEMG, muscle strength and skinfold thickness

Table 2 shows that moderate to high correlations were found between knee extensor AI and iEMG of vastus lateralis during knee extension ($r = 0.75$, $p < 0.001$) and, similarly, between knee extensor AI and iEMG of rectus femoris during knee extension ($r = 0.70$, $p < 0.001$), while a moderate correlation was observed between knee flexor AI and iEMG of semitendinosus during knee flexion ($r = 0.64$, $p < 0.001$). Weak but significant negative correlations were found between thigh skinfold thickness and iEMG of both vastus lateralis ($r = -0.50$, $p = 0.001$) and rectus femoris ($r = -0.42$, $p = 0.009$).

Effects of PRT after 12 weeks of RCT [(Exercise$_{post}$ − Exercise$_{pre}$) vs. (Control$_{post}$ − Control$_{pre}$)]

iEMG

At the pretest, no differences in any iEMG measurements were found between groups (Table 3). In the exercise group, ΔiEMG of vastus lateralis (271 ± 377 vs. 23 ± 249 μV, $p = 0.048$) and rectus femoris (271 ± 299 vs. 47 ± 189 μV, $p = 0.02$) during maximal knee extension and ΔiEMG of semitendinosus (373 ± 133 vs. −57 ± 116 μV, $p = 0.007$) during knee flexion, all increased significantly after the PRT intervention, when compared to the control group (Fig. 3). When compared to the within group pre value, the increases in the exercise

Table 1 Subject characteristics at baseline in all participants and in intervention completers

	All participants	Control	Exercise	*P* value (Control vs. Exercise)
Numbers	38 (13♂/25/♀)	16 (6♂/10/♀)	15 (5♂/10♀)	
Age (years)	48.7 ± 8.8	49.1 ± 8.4	47.7 ± 10.4	n.s.
Height (cm)	169.0 ± 10.6	168.9 ± 12.3	169.8 ± 9.4	n.s.
Weight (kg)	67.7 ± 14.0	66.9 ± 15.2	70.1 ± 14.2	n.s.
EDSS (arbitrary units)	3.8 ± 0.8	3.9 ± 0.9	3.7 ± 0.9	n.s.
Time since diagnose (years)	7.2 ± 5.5	8.1 ± 6.0	6.6 ± 5.9	n.s.
Immunomodullatory treatment (±)	22/16	11/5	7/8	n.s.

Data are given as mean ± SD or numbers

Table 2 Baseline correlations between EMG measures and muscle strength, skinfolds and thigh volume

	VL-iEMG-KE	RF-iEMG-KE	ST-iEMG-KF
KE-AI	0.75*	0.70*	–
KF-AI	–	–	0.64*
Skinfold VL	−0.50*	–	–
Skinfold RF	–	−0.42*	–
TV_{Total}	0.05	0.08	0.09

AI Angular impulse, *iEMG* integrated surface EMG, *KE* knee extension, *KF* knee flexion, *RF* rectus femoris, *ST* semitendinosus, *VL* vastus lateralis

$n = 38$, * $p < 0.001$

group amounted to 36.3 ± 35.9, 40.4 ± 45.2 and 30.8 ± 41.6 % in vastus lateralis, rectus femoris and semitendinosus, respectively. No differences between groups were observed in any measures of co-contraction.

Muscle strength

At the pre test, no differences in muscle strength were found between groups (Table 3). In the exercise group, ΔAI for knee extension (68.0 ± 64.8 vs.−15.5 ± 87.3 N·m·s, $p = 0.007$) and knee flexion (30.4 ± 50.1 vs. −8.5 ± 25.7 N·m·s, $p = 0.003$) improved significantly after the PRT intervention, when compared to the control group (Fig. 3; Table 3). Compared to the within group pre value, the increases in AI in the exercise group for knee extension and knee flexion corresponded to 16.4 ± 4.6 and 24.6 ± 10.6 %, respectively.

Thigh volume and skinfolds

At the pre intervention test, no differences were found between groups for TV_{Total}, $Skinfold_{VL}$ and $Skinfold_{RF}$ (Table 3). A trend was observed towards a significant increase in ΔTV_{Total} in the exercise group, when compared to the control group (0.27 ± 0.05 L vs. 0.13 ± 0.07 L, p = 0.09). However, the exercise group showed a significant time effect in TV_{Total} after the intervention when compared to the pre value ($p < 0.001$), while only a trend ($p = 0.056$) was observed in the control group. No intervention effects were observed in any of the skinfold measurements.

Effects of 12 weeks of follow up in the Exercise group ($Exercise_{post}$ vs. $Exercise_{Follow\ up}$)

It appears from Table 3 that except for AI during knee flexion (change: −22.9 ± 45.9 N·m·s, $p = 0.03$; ~−7.7 ± 15.5 %), no measures of iEMG or muscle strength differed from the post value. The iEMG measures of vastus lateralis and rectus femoris (knee extension) and semitendinosus (knee flexion) remained significantly different from baseline values.

Effects of 12 weeks of PRT in the control group during the follow up period [($Control_{post} - Control_{pre}$) vs. ($Control_{followup} - Control_{post}$)]

Table 3 shows that the effect of PRT in the control group during the 12 weeks follow up period were similar, but less marked and consistent compared to what was found in the exercise group during the first 12 weeks. ΔiEMG of semitendinosus during knee flexion improved significantly after PRT (−57 ± 116 vs. 320 ± 307 μV, $p = 0.04$), while changes in ΔiEMG of vastus lateralis and rectus femoris during knee extension were non-significant (23 ± 249 vs. 195 ± 353 μV, $p = 0.23$ and 47 ± 189 vs. 157 ± 221 μV, $p = 0.11$, respectively). No changes were seen in any measures of co-contraction. ΔAI during knee extension (−15.5 ± 87.3 vs. 53.4 ± 81.5 N·m·s, $p = 0.004$) improved significantly after PRT, while ΔAI during knee flexion (−8.5 ± 25.7 vs. 16.2 ± 28.1 N·m·s, $p = 0.14$) did not change significantly.

Correlations between changes in muscle strength and iEMG

As shown in Fig. 4, moderate but highly significant correlations were found between pooled changes in AI and iEMG of both vastus lateralis ($r = 0.58$, $p < 0.001$) and rectus femoris ($r = 0.61$, $p < 0.001$) during knee extension. A weak but significant correlation was found between pooled changes in AI and iEMG of semitendinosus during knee flexion ($r = 0.42$, $p = 0.02$).

Discussion

This study demonstrated that 12 weeks of supervised PRT of the lower extremities increased maximal neural drive to both knee extensors and knee flexors during maximal isometric contractions. The improved neural drive was maintained after 12 weeks of follow up. The findings could not be completely reproduced in the control group, when this group received PRT during the follow up period. These are important results indicating that the neural plasticity in response to PRT is preserved in MS patients despite a chronic disease in the CNS.

Reduced neural function in MS patients

Since MS is a chronic disease of the CNS, it is not surprising that neural function is affected, as evidenced by

Table 3 Weight, EDSS, muscle strength, iEMG and anthropometric data at pretrial, post-trial and at follow up

	Control ($n = 16$)					Exercise ($n = 15$)				
	Pre	Post	Follow up	Change Post–Pre	Change Follow up-Post	Pre	Post	Follow up	Change Post–Pre	Change Follow up-Post
Weight (kg)	66.9 (15.2)	68.0 (15.7)	68.6 (16.1)	1.1 (1.9)	0.6 (1.3)	70.1 (14.2)	71.1 (14.8)	70.4 (14.2)	1.0 (1.5)	−0.8 (1.5)
EDSS (arbitrary unit)	3.9 (0.9)	4.0 (1.1)	3.8 (1.1)	0.1 (0.4)	−0.2 (0.4)	3.7 (0.9)	3.9 (1.1)	3.7 (1.1)	0.2 (0.5)	−0.2 (0.6)
Muscle strength (Nm*s)										
Knee extensor AI	508.6 (165.6)	493.1 (180.1)	546.5 (236.4)	−15.5 (87.3)	53.4 (81.5)$	514.0 (188.4)	582.0 (193.9)*	555.4 (186.2)	68.0 (64.8)#	−26.5 (51.4)#,$
Knee flexor AI	200.7 (57.8)	192.2 (54.1)	208.4 (70.2)	−8.5 (25.7)	16.2 (28.1)	220.8 (94.1)	251.2 (91.1)*,**	228.4 (76.2)***	30.4 (50.1)#	−22.9 (45.1)#,$
iEMG (μV), knee flexion										
Vastus lateralis	104 (100)	116 (138)	96 (100)	12 (108)	−20 (135)	80 (39)	103.1 (545)	93 (55)	23 (31)	−11 (31)
Rectus femoris	87 (72)	98 (123)	81 (45)	12 (106)	−17 (118)	76 (41)	86.8 (43)	90 (48)	11 (19)	3.5 (36)
Semitendinosus	1183 (465)	1127 (370)	1446 (431)*,***	−57 (116)	320 (307)$,§	1354 (509)	1727.7 (821)*,**	1615 (730)*	373 (133)#	−112 (371)#,$
iEMG (μV), knee extension										
Vastus lateralis	946 (540)	969 (480)	1164 (657)*,***	23 (249)	195 (353)§	1003 (607)	1273.9 (794)*	1212 (649)*	271 (377)#	−61 (363)#,$
Rectus femoris	924 (525)	972 (462)	1129 (452)*,***	47 (189)	157 (221)§	974 (510)	1246 (599)*	1209 (424)*	272 (299)#	−37 (286)#,$
Semitendinosus	126 (46)	164 (159)	144 (57)	38 (147)	−209 (135)	176 (265)	162 (107)	185 (135)	−13 (178)	23 (44)
iEMG (%), co-contraction										
Vastus lateralis	10.9 (6.7)	11.4 (7.9)	8.4 (3.8)	0.6 (2.4)	−3.0 (6.9)	9.4 (4.5)	9.7 (5.9)	8.6 (4.8)	0.3 (6.9)	−1.1 (0.8)
Rectus femoris	10.1 (5.7)	9.3 (5.4)	7.4 (2.7)	−0.8 (6.7)	−1.9 (5.4)	8.9 (4.6)	7.8 (3.9)	7.8 (3.7)	−1.1 (1.7)	0.0 (2.2)
Semitendinosus	11.5 (4.8)	16.0 (17.9)	10.0 (3.3)	4.5 (16.0)	−5.9 (15.9)	11.6 (10.0)	10.1 (5.0)	11.8 (5.0)	−1.4 (8.9)	1.6 (4.1)
Skinfolds (mm)										
Vastus lateralis	22.5 (12.8)	24.0 (10.2)	24.7 (10.9)	1.5 (0.9)	0.6 (0.6)	22.5 (12.8)	21.8 (11.4)	23.3 (12.4)	−0.7 (1.0)	1.5 (0.7)
Rectus femoris	28.5 (10.6)	30.1 (12.1)	30.0 (12.9)	1.7 (1.2)	−0.1 (0.8)	27.9 (10.6)	28.8 (11.1)	28.4 (10.7)	0.9 (1.1)	−0.4 (1.0)
TV_{Total} (L)	6.8 (1.7)	7.0 (1.8)	7.1 (1.9)*,***	0.13 (0.07)	0.15 (0.05)	7.4 (1.4)	7.7 (1.5)*	7.6 (1.4)*	0.27 (0.05)	−0.07 (0.06)#,$

Data are presented as mean ± SD

AI Angular impulse, *EDSS* Expanded Disability Status Score, *iEMG* integrated surface EMG, *TV* tigh volume

Significantly different compared to corresponding change in Control, $significantly different compared to pre to post changes within group, * significantly different compared to pre value within group. ** Significantly different from corresponding control value, *** significantly different from post value within group, §significantly different from post to follow up changes in the exercise group

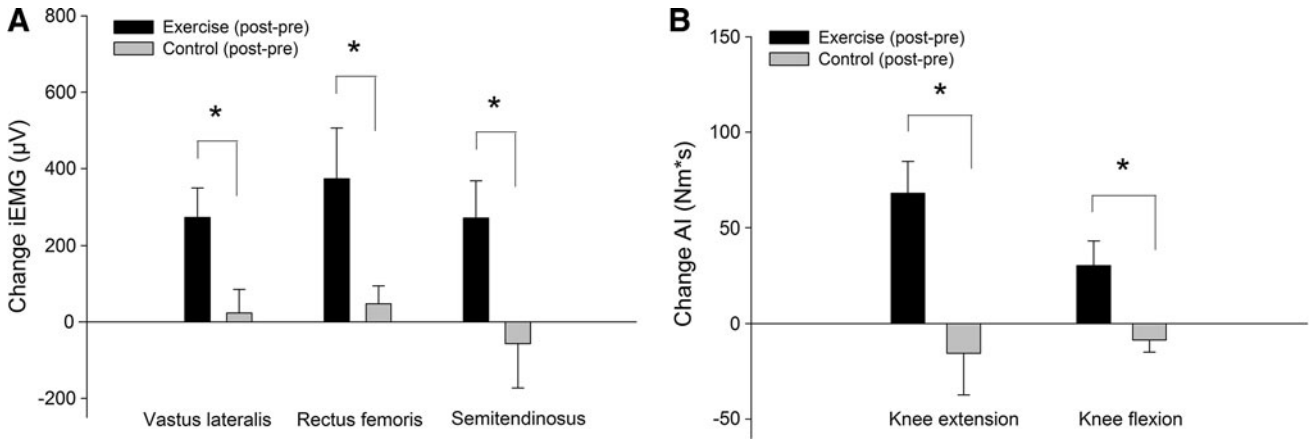

Fig. 3 **a** Effects of 12 weeks of progressive resistance training on changes in integrated EMG (iEMG) of vastus lateralis and rectus femoris during maximal knee extension and semitendinousus during maximal knee flexion. **b** Effects of 12 weeks of progressive resistance training on changes in muscle strength expressed as the angular impulse (AI) during maximal voluntary isometric contractions of the knee extensors and knee flexors in the exercise group (*black bar*) compared to the change in the control group during the trial (*gray bar*). *Asterisk* Change in the exercise group significant different from change in control group, $p < 0.05$

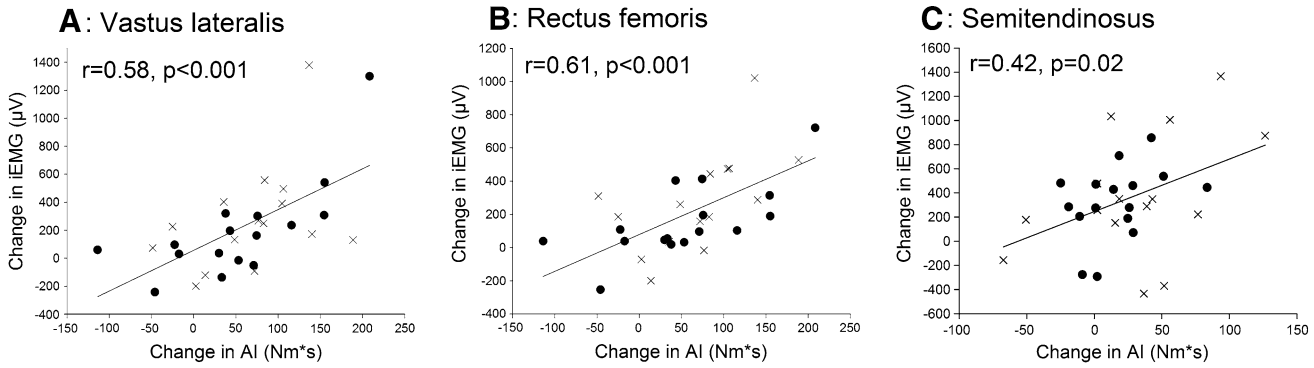

Fig. 4 Correlations between changes in maximal integrated EMG (iEMG) and changes in muscle strength in terms of angular impulse (AI) for **a** vastus lateralis and **b** rectus femoris during maximal knee extension and for **c** semitendinosus during maximal knee flexion. Data from the intervention phase of both study arms ($n = 31$) were pooled and included in the analysis. *Cross symbol* Changes after progressive resistance training in the exercise group ($n = 15$), *filled circle* changes after progressive resistance training in the control group ($n = 16$)

altered neuromuscular recruitment affecting upper motor neurons resulting in impaired force production [9, 24, 25]. Recently, Scott et al. [28] compared surface EMG during maximal and submaximal isometric contractions of the vastus lateralis from moderately impaired MS patients (estimated EDSS: 4–6) to healthy matched controls. Here it was demonstrated, that the neural drive (expressed as the root mean square of the surface EMG recordings) to vastus lateralis was 54 % lower in MS patients than in healthy controls during maximal isometric contractions [28]. Interestingly, our study showed improvements of 36 % in the iEMG, indicating that much of the impairment reported by Scott et al. can be eliminated by PRT. This is also in accordance with our previous report from the present study, showing that PRT improved muscle strength of the knee extensors to a level corresponding to values predicted from normative data in healthy subjects [5]. However, it should be noted that the subcutaneous fat layer was thicker in MS patients in the study by Scott et al. which may have reduced the signal, and thereby overestimated the difference to healthy subjects. After adjusting for the increased subcutaneous layer, the root mean square of the EMG signal was still reduced, but only by approximately 16 % [28]. The impaired neural drive observed in MS patients may be explained by a reduced ability to fully activate motor units of the thigh and lower leg muscles as reported during maximal voluntary contractions in MS patients (47–93 % activation vs. 94–100 % in healthy controls) [8, 23, 24, 29]. Also, decreases in maximal motor unit firing rates have been reported in MS patients [24]. From our study, it cannot be concluded whether or not our participants had impaired neural drive before the intervention, since no matched healthy control subjects were included. However, the neural drive was most likely reduced in our subjects because they 1) had an EDSS score corresponding to the participants in the study by Scott et al. [28] (EDSS

3–5.5 vs. 4–6), where an impairment was shown, and 2) all had a EDSS pyramidal function score ≥2.0, indicating motor neuron damage.

Changes in efferent neural drive

In healthy subjects, intense PRT is a potent stimulus of the nervous system resulting in a number of neural adaptations. These adaptations include changes in motor unit recruitment and/or rate coding, changes in motor neuron excitability, and altered co-activation of antagonist muscles (for references see Aagaard and Thorstensson [3]). Furthermore, Moritani and de Vries showed that neural adaptations dominate in the beginning of a PRT program, as evidenced by an increase in muscle strength without a concomitant hypertrophy until after 3–5 weeks of training [22].

Surface EMG recordings are widely used to measure neural drive to muscles. However, it has been argued that this interpretation may be a simplification since surface EMG is modified by changes in fiber type composition, fiber size, membrane potential, intramuscular ionic concentrations and sodium–potassium pump content, which are all known to influence the single-fiber action potential [11]. Furthermore, the large, fast motor units tend to be more abundant towards the periphery of the muscle close to the skin and any change in their activity may have an exaggerated effect upon surface EMG recordings [20]. Nonetheless, numerous studies have, in accordance with our findings, reported muscle surface EMG to increase significantly after PRT in healthy subjects, particularly during the first 3–4 weeks, and this has been taken as evidence for a change in the neural drive to the muscle[1] (for further references please see Folland et al. [11] and Gabriel et al. [13]). Also, our findings are in accordance with the findings of Fimland et al. [10], who reported that soleus surface EMG activity increased by 40 %, when compared to the control group after 3 weeks of PRT. Despite the impressive results presented by Fimland et al., their applied training frequency of 5 days a week may be difficult to implement in most patients‘ daily life. Furthermore, exercising the same muscle group at such a high training frequency may increase the risk of injury due to insufficient recovery. However, our study provide novel evidence that important neural adaptations can be achieved in the prime movers by only two training sessions per week, lasting approximately 1 h each. This offers an implementable PRT intervention allowing sufficient recovery between bouts.

Methodological considerations

Reproducibility of the EMG measurements in longitudinal studies may be problematic due to difficulties with (1) precise relocation of electrodes, (2) variable impedance of the skin and subcutaneous tissue, as well as (3) changes in muscle morphology [11]. In the present study, we tried to optimize the relocation of the electrodes by carefully mapping the exact electrode positions during the pre-test. Importantly, the consistent iEMG level shown in our control group during the pre and post-test suggests that electrode relocation is a minor problem in the present study. At the pre-test, we demonstrated weak but significant correlations between iEMG and skinfolds of vastus lateralis and rectus femoris. However, our study also showed that neither skinfolds from vastus lateralis and rectus femoris nor whole body fat (data not shown, for details see [5]) changed during the intervention. Consequently, changes in regional or total body fat would not be expected to have influenced the impedance of the subcutaneous tissue after PRT. The present data cannot exclude the possible contribution from morphological adaptations to the training induced increases in iEMG, even though no changes were observed in fiber-type composition (see our earlier paper [7] for details on fiber type issues). Moreover, histochemical analysis of biopsies from vastus lateralis showed an average fiber area increase of 8 % and a preferential type II fiber hypertrophy after PRT (14 % increase in type II fiber cross-sectional area) [7]. Hypertrophy per se would not be expected to increase the iEMG during maximal isometric contractions, but the fact that a preferential type II fiber hypertrophy was observed could have influenced the iEMG, albeit to a minor degree only [1].

Another methodological consideration concerns possible EMG crosstalk between the knee extensors and knee flexors. However, it has been shown that only negligible amounts of EMG crosstalk were observed during maximal contractions in a study applying a similar bipolar electrode configuration [2]. A final important technical remark concerns the applied EMG sampling frequency, which was only 100 Hz, i.e., considerably below the ≥1,000 Hz applied in other studies [1, 28]. As a consequence of the low sampling frequency, we only evaluated iEMG after low-pass filtering, which is virtually unaffected by the sampling frequency. Moreover, we tried to downsample data obtained at 1500 Hz from another study to 100 Hz to see if iEMG was affected, which was not the case. However, the low sampling frequency did not allow us to do a frequency analysis of the signal.

The present study showed correlations between changes in muscle strength (AI) and iEMG (see Fig. 4), but findings also showed that absolute AI changes during knee extension were larger than AI changes during knee flexion, while at the same time the changes in iEMG of semitendinosus during knee flexion, were larger than seen in rectus femoris during knee extension (see Table 3). However, it cannot be expected, that the relative changes in muscle strength and iEMG should be equal. Firstly, the applied measures of

muscle strength are comprised by activity in several muscles acting together on the knee joint, whereas the iEMG measures reflect only one of these muscles (actually only part of the muscle). Secondly, there is not a linear relationship between muscle activation frequency and muscle strength. If a motor unit is activated with an increasing frequency the strength may increase with gradually smaller improvements as the frequency approaches tetanic frequency. However, the action potentials, which make up the EMG signal, may increase linearly with increasing frequency.

In addition to the training-induced adaptations to neural drive evidenced by the increase in maximal EMG it is also known that more acute regulation of the central drive and motor pathway excitability can occur during fatigue. These acute fatigue-induced changes may be particularly prominent in MS patients, as evidenced by superimposed stimulation [30] and motor evoked potentials [31]. Although we have not measured excitability of motor pathways in this study, we have attempted to avoid possible effects of fatigue by measuring strength only briefly in rested muscles.

Reproducibility of the training effects in the control group

After 12 weeks of PRT in the control group, only some of the changes seen in the exercise group after PRT could be reproduced to a degree, where the changes were statistically significant. Thus, iEMG of semitendinosus during knee flexion improved significantly after PRT, whereas iEMG of vastus lateralis and rectus femoris during knee extension did not show significant changes, despite significant increases in AI during maximal knee extension. Although these discrepancies complicate the interpretation of the data somewhat, we emphasize that the main pattern of changes in mean values of the control group after the PRT intervention was always towards improvement (Table 3). A post hoc sample size calculation (power: 80 %, $p \leq 0.05$) based on the slightly higher variation obtained in the control group for the iEMG of rectus femoris during knee extension, showed that $n = 28$ participants were needed, to show a significant group difference of the changes during the follow period, suggesting that the study is slightly underpowered increasing the risk of a type 2 error. A similar pattern of training effects in the control group is also supported by the fact that all iEMG follow up values differed from both pre and post values within group, when tested with a two-way ANOVA over time. This was also the case when post measurements of the exercise group were compared to the within group pre values. Also, our correlation analysis showed a resembling pattern in the control group, when changes in iEMG were plotted against changes in muscle strength (Fig. 4). However, since the control group generally showed smaller improvements in AI after PRT than was observed in the exercise group, which was paralleled by smaller improvements in iEMG, it must be concluded that for some reason the training effects were less marked and consistent in this group. We have no clear explanation for the generally minor improvements observed in the control group. One possibility could be that the participants of the control group might have been less motivated for intense training after being randomized to a control group for a 12-week "waiting period", despite our attempts to make all participants feel equally important.

Limitations

Only the best functioning leg was tested and it can, therefore, not be stated whether the most affected leg would have shown the same pattern of adaptations. One could argue that it would be easier to improve the functional level of the most affected leg due to a lower starting level before a PRT intervention leaving more room for improvement. In contrast, more irreversible neuronal damage could be present in the most affected leg limiting the adaptability. Secondly, the participants represent a selected group of moderately impaired relapsing-remitting MS patients. It seems reasonable to expect that more mildly affected MS patients (EDSS score between 0 and 3) would be able to tolerate and benefit from PRT in the same manner as the moderate affected group included in this study. However, it is more questionable whether the results can be generalized to the more severely impaired MS patients (EDSS > 5.5). Finally, the participants and some of the assessors were not blinded to the intervention.

Conclusions

Twelve weeks of intense progressive resistance training of the lower extremities improves the maximal neural drive in patients with multiple sclerosis, with effects persisting after 12 weeks of self-guided physical activity.

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